Evidencia Situación problema Series Tiempo - Capital Analyst

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Series de tiempo

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Colab 1M
https://colab.research.google.com/drive/1mK1NB6LQI4B7_16CxNQKqH7adAbznH7t?usp=sharing

Colab 30M
https://colab.research.google.com/drive/12rW3UGxidzOZXPfLc5ftyINbJB_CMm2_?usp=sharing

Screenshot

SYMBOL COMPANY	QTY	PRICE PAID	LAST PRICE	DAY'S CHANGE	MARKET VALUE	% PROFIT/LOSS	
JNJ Johnson & ...	100	156.25	156.50	2.04	$15,649.50	24.50 (0.16%)	TRADE
PEP PepsiCo Inc	-100	134.12	133.96	0.41	$13,396.00	16.00 (0.12%)	TRADE
AAPL Apple Inc	-100	203.89	204.51	-8.81	$20,451.10	-62.10 (-0.30%)	TRADE

Prompt
https://grok.com/chat/9d3948fb-f3ed-4515-8fdc-b907d69f7670

Github
https://github.com/A00573236/Evidencia-Situaci-n-Problema

Interpretación (1M)

Data description

- **Sample Size**: Approximately 292 observations per stock, aligned to the shortest series length.
- **Time Frame**: Recent data, likely spanning a short period in April 2025
- **Frequency**: 1-minute intervals.

Analysis and results

Unit root tests

Unit root tests, including Augmented Dickey-Fuller and Kwiatkowski-Phillips-Schmidt-Shin, were conducted to determine whether the price series are stationary or non-stationary.

- **AAPL**:

    - **ADF**: Statistic = -1.1557, p-value = 0.6923 (> 0.05). Fail to reject null; non-stationary.
    - **KPSS**: Statistic = 1.1983, p-value = 0.0100 (< 0.05). Reject null; non-stationary.
    - **Interpretation**: AAPL's price series exhibits non-stationary behavior, likely following a trend or random walk, suitable for trend-following strategies.
- **JNJ**:
    - **ADF**: Statistic = -3.1643, p-value = 0.0221 (< 0.05). Reject null; stationary.
    - **KPSS**: Statistic = 0.2890, p-value = 0.1000 (> 0.05). Fail to reject null; stationary.
    - **Interpretation**: JNJ's price series is stationary, indicating mean-reverting behavior, which supports strategies anticipating price reversion to a stable mean.
- **PEP**:
    - **ADF**: Statistic = -2.5899, p-value = 0.0951 (> 0.05). Fail to reject null; non-stationary.
    - **KPSS**: Statistic = 1.4906, p-value = 0.0100 (< 0.05). Reject null; non-stationary.
    - **Interpretation**: PEP is non-stationary, suggesting potential trending behavior similar to AAPL.

The results remain valid. JNJ's stationarity is atypical for stock prices, possibly due to the short time frame or specific data characteristics.

Cointegration tests

Johansen cointegration tests were performed to identify long-run equilibrium relationships between stock pairs, which can enable pairs trading strategies based on spread reversion.

- **AAPL and JNJ**:
    - Trace statistic: [29.54, 1.41]
    - Critical values (95%): [15.49, 3.84]
    - **Result**: r = 0: 29.54 > 15.49 (cointegration at 95%); r = 1: 1.41 < 3.84 (no further cointegration).
    - **Conclusion**: AAPL and JNJ are cointegrated, sharing a long-run equilibrium. Deviations in their spread (AAPL - β * JNJ) are expected to revert.
- **AAPL and PEP**:
    - Trace statistic: [10.43, 2.00]
    - Critical values (95%): [15.49, 3.84]
    - **Result**: r = 0: 10.43 < 15.49; r = 1: 2.00 < 3.84. No cointegration.
    - **Conclusion**: AAPL and PEP lack a long-run relationship, making pairs trading less viable.
- **JNJ and PEP**:

    - Trace statistic: [38.92, 6.83]
    - Critical values (95%): [15.49, 3.84]
    - **Result**: r = 0: 38.92 > 15.49; r = 1: 6.83 > 3.84. Cointegration at both ranks.
    - **Conclusion**: JNJ and PEP are strongly cointegrated, supporting pairs trading based on spread reversion.

Cointegration between AAPL-JNJ and JNJ-PEP suggests opportunities for pairs trading, where one stock is bought and the other sold when their spread deviates from its mean.

ARIMA price forecasts

ARIMA models were fitted to forecast prices over a 30-period horizon. The best model orders were selected using the Akaike Information Criterion.

- **AAPL**:
    - **Model**: ARIMA(0,1,1), AIC = -1.88
    - **Last Observed Price**: $209.75
    - **Average Forecasted Price**: $209.75
    - **Change**: $0.00
    - **Trend**: Flat, leaning downward
    - **95% Confidence Interval (period 30)**: [$208.09, $211.40]
    - **Interpretation**: The flat forecast indicates no significant price growth, suggesting a neutral to bearish outlook. The narrow confidence interval reflects moderate forecast uncertainty.
- **JNJ**:
    - **Model**: ARIMA(3,0,3), AIC = 95.22
    - **Last Observed Price**: $156.25
    - **Average Forecasted Price**: $155.91
    - **Change**: -$0.34
    - **Trend**: Downward
    - **95% Confidence Interval (period 30)**: [$154.89, $156.53]
    - **Interpretation**: A slight decline is expected, but the small magnitude and narrow confidence interval align with JNJ's stationary, stable behavior.
- **PEP**:
    - **Model**: ARIMA(2,1,3), AIC = -535.38
    - **Last Observed Price**: $135.00
    - **Average Forecasted Price**: $134.94
    - **Change**: -$0.06
    - **Trend**: Downward
    - **95% Confidence Interval (period 30)**: [$134.23, $135.64]
    - **Interpretation**: The minimal decline and narrow confidence interval suggest a stable but slightly bearish outlook, consistent with PEP's non-stationary nature.

ARIMA fitting indicates potential model instability, particularly for JNJ and PEP's complex orders. Simpler models could improve robustness but were not tested here.

GARCH volatility forecasts

GARCH(1,1) models were applied to log returns to forecast annualized volatility over 30 periods, providing insights into expected price fluctuations.

- **AAPL**:
  - **Last 30-day Volatility**: 0.47%
  - **Average Forecasted Volatility**: 1.23%
  - **Change**: +0.76%
  - **Trend**: Increasing
  - **First 5 Periods**: [0.87%, 0.90%, 0.93%, 0.96%, 0.99%]
  - **Interpretation**: Rising volatility suggests increased risk and potential for larger price swings, which could impact trading strategies.
- **JNJ**:
  - **Last 30-day Volatility**: 4.80%
  - **Average Forecasted Volatility**: 3.23%
  - **Change**: -1.57%
  - **Trend**: Decreasing
  - **First 5 Periods**: [3.22%, 3.24%, 3.23%, 3.26%, 3.26%]
  - **Interpretation**: Declining volatility indicates stabilizing price movements, reducing risk and supporting mean-reversion strategies.
- **PEP**:
  - **Last 30-day Volatility**: 3.36%
  - **Average Forecasted Volatility**: 4.94%
  - **Change**: +1.58%
  - **Trend**: Increasing
  - **First 5 Periods**: [4.86%, 4.89%, 4.88%, 4.91%, 4.93%]
  - **Interpretation**: Increasing volatility suggests heightened risk, potentially amplifying price movements.

GARCH scaling warnings highlight low return scales, suggesting rescaling could improve estimation. The current results are interpretable, though AAPL's low observed volatility appears unusually low for annualized 1-minute data, warranting verification.

Decisions made

AAPL (Apple)

- **Decision:** Short
- **Rationale**:

    - **Price Trend**: The flat forecast ($209.75, no change) indicates limited upside potential, supporting a bearish stance.
    - **Volatility**: Increasing volatility (0.47% to 1.23%) suggests higher risk, making holding less attractive and shorting viable for active traders.
    - **Cointegration**: Cointegration with JNJ enables pairs trading. If AAPL is overvalued relative to JNJ (spread > 0), shorting AAPL while buying JNJ could be profitable.
    - **Stationarity**: Non-stationary behavior aligns with trend-following, but the flat forecast leans bearish.

JNJ (Johnson & Johnson)

- **Decision:** Buy
- **Rationale**:
    - **Price Trend**: The modest decline ($156.25 to $155.91, -0.34) is minimal, and stationarity suggests potential reversion to the mean.
    - **Volatility**: Decreasing volatility (4.80% to 3.23%) indicates lower risk, enhancing JNJ's appeal as a stable investment.
    - **Cointegration**: Cointegration with AAPL and PEP supports pairs trading. If JNJ is undervalued (spread < 0), buying JNJ is advantageous.
    - **Stationarity**: Stationary behavior favors mean-reversion strategies, as JNJ is likely to return to its historical mean.

PEP (PepsiCo)

- **Decision:** Short
- **Rationale**:
    - **Price Trend**: The slight decline ($135.00 to $134.94, -0.06) suggests bearish momentum, though limited in magnitude.
    - **Volatility**: Increasing volatility (3.36% to 4.94%) heightens risk, favoring short-term bearish strategies.
    - **Cointegration**: Cointegration with JNJ supports pairs trading. If PEP is overvalued (spread > 0), shorting PEP while buying JNJ is recommended.
    - **Stationarity**: Non-stationary behavior aligns with trend-following, and the downward forecast supports a bearish outlook.

Results

- **AAPL**: The ARIMA forecast predicted a flat price of $209.75 for AAPL over a 30-minute horizon, but the actual price is $204.51, below the forecasted range of $208.09–$211.40, despite a slight increase from the price paid of $203.89; the intraday drop of -$8.81 suggests unmodeled market events or volatility (forecasted to rise to 1.23%) over the elapsed time, causing the discrepancy, so with the short position currently profitable (-42.10%), it's recommended to hold the short or cover if the price nears $208.09 to lock in gains while setting a stop-loss at that level.

- **JNJ**: For JNJ, the ARIMA model forecasted a slight decline to $155.91 from $156.25, aligning closely with the actual price of $156.50 within the confidence interval of $154.89–$156.53, though the +$2.04 intraday gain indicates positive market sentiment not captured by the short-term downward trend forecast; given the long position's +24.50% profit and decreasing volatility (forecasted at 3.23%), holding or adding to the position is advised, with a stop-loss at $154.89 to manage risk.
- **PEP**: The ARIMA forecast for PEP anticipated a minor drop to $134.94 from $134.12, which was directionally correct but underestimated the actual decline to $133.96, falling below the confidence interval of $134.23–$135.64, likely due to unmodeled factors over time and increasing volatility (forecasted at 4.94%) reflected in the +$0.41 intraday gain; the short position's +16.00% profit supports holding the short or covering if the price approaches $134.23, with a stop-loss set at that level to guard against volatility-driven reversals.

Own interpretations

The analysis of AAPL, JNJ, and PEP stock prices over a short, high-frequency period reveals distinct behaviors. AAPL and PEP are non-stationary, with flat to bearish ARIMA forecasts and rising volatility, supporting short positions. Actual prices fell below forecasts, indicating unmodeled volatility or market events. JNJ, however, is stationary, with a slightly declining forecast and decreasing volatility, favoring long positions due to mean-reverting tendencies.

Cointegration tests show strong long-run relationships between JNJ-AAPL and JNJ-PEP, enabling pairs trading strategies. Shorting AAPL or PEP when overvalued relative to JNJ, while buying JNJ, leverages spread reversion. AAPL and PEP lack cointegration, limiting their pairing. The models show some instability, with forecast errors suggesting caution and tight stop-losses.

Overall, JNJ's stability makes it ideal for long positions or pairs trading, while AAPL and PEP suit short-term bearish strategies. Volatility and model limitations require careful monitoring and risk management, but cointegration offers a solid framework for exploiting price relationships in this short-term, dynamic market.

Interpretación (30M)

Data description

- **Sample Size**: Approximately 121 observations per stock.
- **Time Frame**: Recent data, likely spanning a short period in April 2025
- **Frequency**: 30-minute intervals.

Unit root tests

AAPL

- **ADF Test**: Statistic = -1.7157, p-value = 0.4231 (fail to reject null: non-stationary). Critical Values: 1% (-3.488), 5% (-2.887), 10% (-2.580).
- **KPSS Test**: Statistic = 1.2921, p-value = 0.0100 (reject null: non-stationary). Critical Values: 10% (0.347), 5% (0.463), 2.5% (0.574), 1% (0.739).
- **Conclusion**: AAPL is non-stationary (I(1)), consistent with stock price behavior.

JNJ

- **ADF Test**: Statistic = -1.4093, p-value = 0.5778 (fail to reject null: non-stationary). Critical Values: 1% (-3.487), 5% (-2.886), 10% (-2.580).
- **KPSS Test**: Statistic = 1.4096, p-value = 0.0100 (reject null: non-stationary). Critical Values: 10% (0.347), 5% (0.463), 2.5% (0.574), 1% (0.739).
- **Conclusion**: JNJ is non-stationary (I(1)).

PEP

- **ADF Test**: Statistic = -2.9551, p-value = 0.0393 (reject null: stationary). Critical Values: 1% (-3.487), 5% (-2.886), 10% (-2.580).
- **KPSS Test**: Statistic = 0.3170, p-value = 0.1000 (fail to reject null: stationary). Critical Values: 10% (0.347), 5% (0.463), 2.5% (0.574), 1% (0.739).
- **Conclusion**: PEP appears stationary, which is atypical for stock prices (usually I(1)). This may reflect a short sample size or data anomaly, warranting caution in cointegration analysis.

Cointegration tests

AAPL and JNJ

- **Trace Statistics**: [19.05, 2.77]
- **Critical Values (90%, 95%, 99%)**: [[13.4294, 15.4943, 19.9349], [2.7055, 3.8415, 6.6349]]
- **Results**: r = 0: Trace (19.05) > 95% critical value (15.49) → Cointegration exists. r = 1: Trace (2.77) ≤ 95% critical value (3.84) → No further cointegration.
- **Conclusion**: AAPL and JNJ are cointegrated, suggesting price deviations may revert.

AAPL and PEP

- **Trace Statistics**: [21.84, 4.59]
- **Critical Values (90%, 95%, 99%)**: [[13.4294, 15.4943, 19.9349], [2.7055, 3.8415, 6.6349]]

- **Results**: r = 0: Trace (21.84) > 95% critical value (15.49) → Cointegration exists. r = 1: Trace (4.59) > 95% critical value (3.84) → Additional cointegration.
- **Conclusion**: Cointegration exists, but PEP's stationarity undermines reliability.

JNJ and PEP

- **Trace Statistics**: [16.46, 5.17]
- **Critical Values (90%, 95%, 99%)**: [[13.4294, 15.4943, 19.9349], [2.7055, 3.8415, 6.6349]]
- **Results**: r = 0: Trace (16.46) > 95% critical value (15.49) → Cointegration exists. r = 1: Trace (5.17) > 95% critical value (3.84) → Additional cointegration.
- **Conclusion**: Cointegration exists, but PEP's stationarity suggests caution.

PEP's stationarity questions the validity of cointegration results, as I(1) series are required. AAPL-JNJ cointegration is most reliable.

ARIMA price forecasts

AAPL

- **Model**: ARIMA(3,1,3), AIC = 245.51
- **Forecast**:
  - Last Value: 212.59
  - Average Forecast: 211.55
  - Change: -1.04
  - Trend: Downward
  - 95% CI (period 30): [208.05, 214.71]
  - Next 5 Periods: [212.56, 212.33, 211.89, 212.01, 211.77]
- **Interpretation**: A modest decline is expected.

JNJ

- **Model**: ARIMA(2,1,3), AIC = 135.85
- **Forecast**:
  - Last Value: 156.04
  - Average Forecast: 156.03
  - Change: -0.01
  - Trend: Downward (nearly flat)
  - 95% CI (period 30): [153.47, 158.67]
  - Next 5 Periods: [155.97, 156.18, 155.97, 155.93, 156.17]
- **Interpretation**: Prices are stable with minimal change.

PEP

- **Model**: ARIMA(2,1,0), AIC = 161.74

- **Forecast**:
  - Last Value: 135.00
  - Average Forecast: 135.12
  - Change: +0.12
  - Trend: Upward
  - 95% CI (period 30): [131.46, 138.79]
  - Next 5 Periods: [135.06, 135.16, 135.13, 135.12, 135.12]
- **Interpretation**: A slight increase is expected.

ARIMA models for AAPL and JNJ show convergence warnings and non-invertible MA parameters, suggesting potential forecast instability. Modest changes indicate cautious interpretation.

GARCH Volatility Forecasts

AAPL

- **Model**: mu = 0.0206 (p = 0.391), omega = 0.0464 (p = 0.007), alpha[1] = 0.7534 (p = 0.086), beta[1] = 0.0239 (p = 0.592). AIC = 58.3119.
- **Forecast**:
  - Last 30-day Volatility: 6.63%
  - Average Forecasted Volatility: 7.04%
  - Change: +0.41%
  - Trend: Increasing
  - Next 5 Periods: [4.07%, 5.00%, 5.64%, 5.95%, 6.30%]
- **Interpretation**: Higher volatility increases risk and trading potential.

JNJ

- **Model**: mu = 0.0258 (p = 0.178), omega = 0.0385 (p = 0.004), alpha[1] = 0.4240 (p = 0.039), beta[1] = 0.1319 (p = 0.106). AIC = 24.8808.
- **Forecast**:
  - Last 30-day Volatility: 6.19%
  - Average Forecasted Volatility: 4.61%
  - Change: -1.58%
  - Trend: Decreasing
  - Next 5 Periods: [4.13%, 4.31%, 4.52%, 4.56%, 4.58%]
- **Interpretation**: Lower volatility suggests stability.

PEP

- **Model**: mu = 0.0187 (p = 0.849), omega = 0.0888 (p = 0.189), alpha[1] = 0.5080 (p = 0.708), beta[1] = 0.0412 (p = 0.725). AIC = 93.0087.
- **Forecast**:
  - Last 30-day Volatility: 6.56%

- Average Forecasted Volatility: 7.01%
    - Change: +0.45%
    - Trend: Increasing
    - Next 5 Periods: [6.70%, 6.83%, 6.81%, 6.86%, 7.02%]
- **Interpretation**: Higher volatility increases risk and opportunity.

JNJ's GARCH model shows a data scale warning, suggesting rescaling log returns for better convergence. Results remain usable.

Decisions made

AAPL

- **Desicion:** Sell or Short
- **Rationale**:
    - ARIMA predicts a downward trend (change: -1.04, from 212.59 to 211.55), suggesting price declines.
    - GARCH indicates increasing volatility (7.04%, +0.41%), amplifying potential gains from shorting.
    - Cointegration with JNJ supports pairs trading (buy JNJ, short AAPL if AAPL overperforms).
- **Considerations**: The 95% CI (208.05–214.71) includes upside potential, and ARIMA convergence issues suggest caution. Monitor price trends and pair dynamics.

JNJ

- **Desicion:** Hold
- **Rationale**:
    - ARIMA predicts a nearly flat trend (change: -0.01, from 156.04 to 156.03), indicating stability.
    - GARCH shows decreasing volatility (4.61%, -1.58%), reducing risk but limiting short-term gains.
    - Cointegration with AAPL supports pairs trading (buy JNJ, short AAPL if divergence occurs).
- **Considerations**: The 95% CI (153.47–158.67) allows slight upside. ARIMA warnings suggest verifying stability.

PEP

- **Desicion:** Short
- **Rationale**:
    - ARIMA predicts an upward trend (change: +0.12, from 135.00 to 135.12), suggesting modest gains.
    - GARCH indicates increasing volatility (7.01%, +0.45%), enhancing potential returns.

    - Cointegration with AAPL and JNJ is questionable due to PEP's stationarity.
- **Considerations**: PEP's stationarity (unusual for stock prices) undermines cointegration and ARIMA reliability. The 95% CI (131.46–138.79) includes downside risk. Confirm trend with additional data.

Results

- **AAPL:** The ARIMA model predicted a downward price trend for AAPL, forecasting a decline from 212.59 to 211.55 (change: -1.04) over 30 periods, with GARCH indicating increasing volatility (from 6.63% to 7.04%), supporting a sell or short recommendation; however, the actual last price of 204.51 (down from 203.89 paid, with a day's change of -8.81) reflects a larger drop than predicted, yet your short position incurred a -62.10 loss (-0.30%) due to a slight price increase from 203.89 to 204.51, likely due to a significant earlier decline followed by a short-term rebound, possibly driven by unmodeled market events like tech sector news.
- **JNJ:** The ARIMA model forecasted a nearly flat trend for JNJ, with a slight decrease from 156.04 to 156.03 (change: -0.01), and GARCH predicted decreasing volatility (from 6.19% to 4.61%), suggesting a hold or sell stance; in reality, the last price of 156.50 (up from 156.25 paid, with a day's change of +2.04) resulted in a +24.50 profit (+0.16%) for your long position, indicating a slight upward movement within the 95% CI (153.47–158.67), likely influenced by unaccounted factors such as healthcare sector gains or JNJ-specific news.
- **PEP:** The ARIMA model predicted an upward trend for PEP, with a rise from 135.00 to 135.12 (change: +0.12), and GARCH forecasted increasing volatility (from 6.56% to 7.01%), supporting a buy recommendation; however, the actual last price of 133.96 (down from 134.12 paid, with a day's change of +0.41) led to a +16.00 profit (+0.12%) for your short position, reflecting a decline of 1.04 from the last observed price, possibly due to PEP's atypical stationarity affecting model accuracy or external factors like market corrections overriding the forecasted upward trend.

Own interpretations

The analysis of AAPL, JNJ, and PEP reveals distinct behaviors in their price and volatility dynamics, with implications for trading strategies. For AAPL, the ARIMA model predicted a modest downward trend, supported by increasing volatility per the GARCH model, justifying a sell or short decision. However, the actual price dropped significantly to 204.51, yet a short position incurred a loss due to a slight rebound from 203.89. This discrepancy suggests that while the downward trend was correctly anticipated, unmodeled market events, possibly tech sector volatility or macroeconomic news caused a larger initial drop followed by a partial recovery. The cointegration with JNJ further supports pairs trading opportunities, but ARIMA convergence issues and the wide 95% CI highlight the need for caution and real-time monitoring.

JNJ exhibited stability, with ARIMA forecasting a nearly flat trend and GARCH indicating decreasing volatility, aligning with a hold recommendation. The actual price rose slightly to 156.50, yielding a small profit on a long position, consistent with the 95% CI. This outcome reflects JNJ's resilience, likely driven by steady demand in the healthcare sector or company-specific developments not captured in the models. The cointegration with AAPL strengthens the case for pairs trading, but ARIMA warnings and the GARCH data scale issue suggest potential model instability. JNJ's low volatility and stable price make it a safer bet in volatile markets, though limited upside caps short-term gains.

PEP's analysis is complicated by its unexpected stationarity, which contradicts typical stock price behavior (I(1)). The ARIMA model predicted a slight upward trend, and GARCH forecasted rising volatility, suggesting a buy. However, the actual price fell to 133.96, yet a short position profited due to a decline from 134.12. This misalignment likely stems from PEP's stationarity undermining cointegration with AAPL and JNJ, rendering ARIMA less reliable. External factors, such as consumer goods sector corrections, may have driven the price drop. The wide 95% CI and model limitations highlight the need for additional data to confirm trends, making PEP the riskiest of the three for trading decisions.